

EXPERIMENT-4

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Semester: 5
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AIM:

To understand the concept of Static Routing, configure static routes using multiple routers in Cisco Packet Tracer, and verify communication between hosts located in different networks.

OBJECTIVES:

- To understand the concept of routing in computer networks.
- To study the working of routing tables.
- To configure static routes between multiple routers.
- To configure IP addresses and default gateways on hosts.
- To establish communication between different LANs using routers.
- To compare Static Routing and Dynamic Routing.
- To verify successful communication using the Ping command and Simulation Mode.

SOFTWARE REQUIREMENTS:

(a) Software:

- Cisco Packet Tracer
- Windows Operating System

(b) Hardware (Virtual):

- Cisco 2911 Routers
- Cisco 2960-24TT Switches

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- PC-PT
- Copper Straight-Through Cable
- Copper Cross-Over Cable

THEORY: (i) Routing

Routing is the process of selecting the best path for forwarding data packets from a source network to a destination network. Routers examine the destination IP address of incoming packets and determine the most appropriate path using information stored in a routing table.

If the source and destination devices belong to the same network, communication occurs directly through a switch. However, when devices belong to different networks, a router is required to forward packets between them.

(ii) Router

A Router is a Layer-3 networking device that connects two or more different networks. It forwards packets based on IP addresses and maintains a Routing Table containing information about reachable networks and the next-hop addresses.

Each router interface belongs to a separate network and acts as the Default Gateway for devices connected to that network.

Functions of a Router:

- Connects multiple LANs.
 - Uses IP addresses for forwarding.
 - Maintains a Routing Table.
 - Separates broadcast domains.
 - Provides communication between different networks.
 - Acts as the Default Gateway.
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(iii) Routing Table

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A Routing Table is a database maintained by a router that stores information about destination networks and the best path to reach them.

Each entry generally contains:

- Destination Network
- Subnet Mask
- Next-Hop Address
- Exit Interface

Whenever a packet arrives, the router searches the routing table and forwards the packet through the appropriate interface.

(iv) Static Routing

Static Routing is a routing technique in which routes are manually configured by the network administrator.

The administrator explicitly specifies the destination network and the next-hop IP address. Routers do not exchange routing information automatically, making static routing suitable for small and stable networks.

Advantages

- Easy to configure for small networks.
- More secure because no routing advertisements are exchanged.
- Low CPU utilization.
- Low bandwidth consumption.
- Predictable routing paths.

Disadvantages

- Manual configuration required.
- Difficult to maintain in large networks.
- No automatic updates when the topology changes.

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- Less scalable.
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(v) Dynamic Routing

Dynamic Routing is a routing technique in which routers automatically exchange routing information using routing protocols such as RIP, OSPF, EIGRP, and BGP.

Routers automatically discover new networks and update their routing tables whenever the network topology changes.

Advantages

- Automatic route learning.
- Scalable for large networks.
- Automatically adapts to failures.
- Selects optimal paths.

Disadvantages

- Higher CPU utilization.
 - Higher memory usage.
 - Additional bandwidth consumed by routing updates.
 - More complex configuration.
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Difference Between Static Routing and Dynamic Routing:

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Route Updates	Manual	Automatic
Scalability	Small Networks	Large Networks
CPU Usage	Low	High

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Memory Usage	Low	High
Bandwidth Usage	Low	Higher
Security	More Secure	Less Secure
Maintenance	Difficult	Easier
Network Changes	Manual Update	Automatic Learning
Best Suitable For	Small Networks	Medium & Large Networks

STEPS/IMPLEMENTATION:

Part-I: Static Routing using Two Routers

1. Place two Cisco 2911 Routers.
2. Place two PCs.
3. Connect Router0 and Router1 using a Copper Cross-Over cable.
4. Connect each PC to its respective router.
5. Assign IP addresses to all interfaces.
6. Configure Default Gateway on both PCs.
7. Initially verify communication (communication fails because no routes exist).
8. Configure Static Routes on both routers.
9. Verify communication using Ping and Simple PDU.

Part-II: Static Routing using Three Routers

1. Place three Cisco 2911 Routers.
2. Connect the routers using Cross-Over cables.
3. Connect one switch to each router.
4. Connect two PCs to every switch.

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5. Assign IP addresses to all PCs and router interfaces.
6. Configure Default Gateway on each PC.
7. Configure Static Routes on all three routers.
8. Verify end-to-end communication between different LANs.

PROCEDURE:

1. Part-I (Two Router Topology)

- Configure Router0 interface with **10.0.0.1** and **11.0.0.1**.
- Configure Router1 interface with **12.0.0.1** and **11.0.0.2**.
- Configure PC0 as **10.0.0.10**.
- Configure PC1 as **12.0.0.10**.
- Configure Default Gateway:
 - PC0 → **10.0.0.1**
 - PC1 → **12.0.0.1**
- Configure Static Routes using the next-hop IP addresses.
- Verify connectivity using Ping.

2. Part-II (Three Router Topology)

Configure three LANs.

LAN-1

- Network: **192.168.1.0/24**
- Gateway: **192.168.1.4**

PC0 → 192.168.1.2

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PC1 → 192.168.1.3

LAN-2

- Network: **192.168.2.0/24**
- Gateway: **192.168.2.4**

PC2 → 192.168.2.2

PC3 → 192.168.2.3

LAN-3

- Network: **192.168.3.0/24**
- Gateway: **192.168.3.4**

PC4 → 192.168.3.2

PC5 → 192.168.3.3

Configure inter-router links.

Router1 ↔ Router2

192.168.4.3

192.168.4.4

Router2 ↔ Router3

192.168.5.2

192.168.5.3

Configure static routes on every router and verify communication.

INPUT/OUTPUT DETAILS:

(a) Devices Used

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- Cisco 2911 Router
- Cisco 2960-24TT Switch
- PC-PT

(b) Cable Types

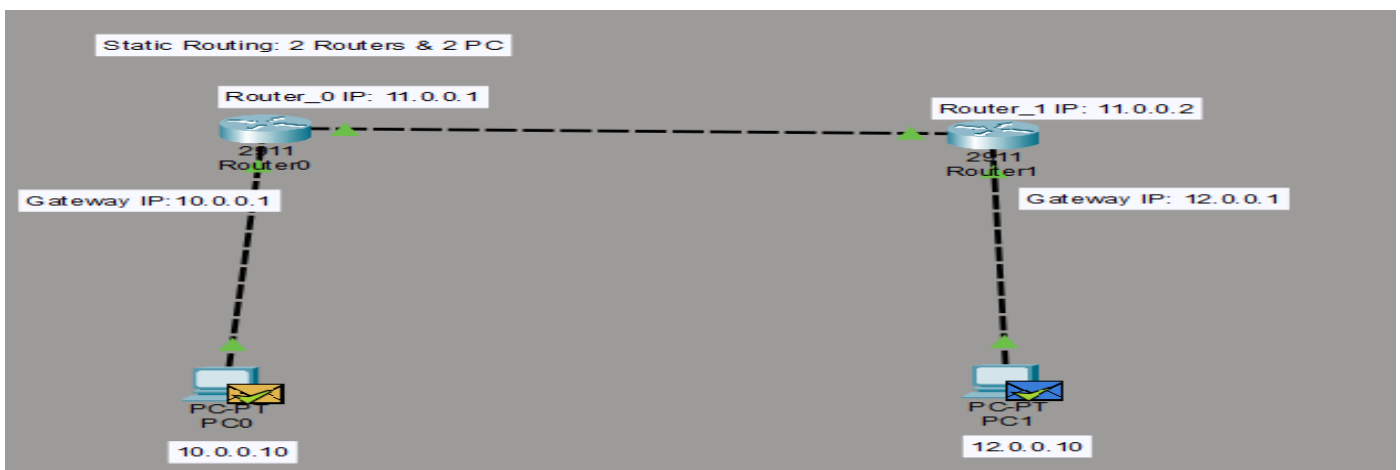
- Copper Straight-Through
- Copper Cross-Over

(c) Cisco IOS Commands Used

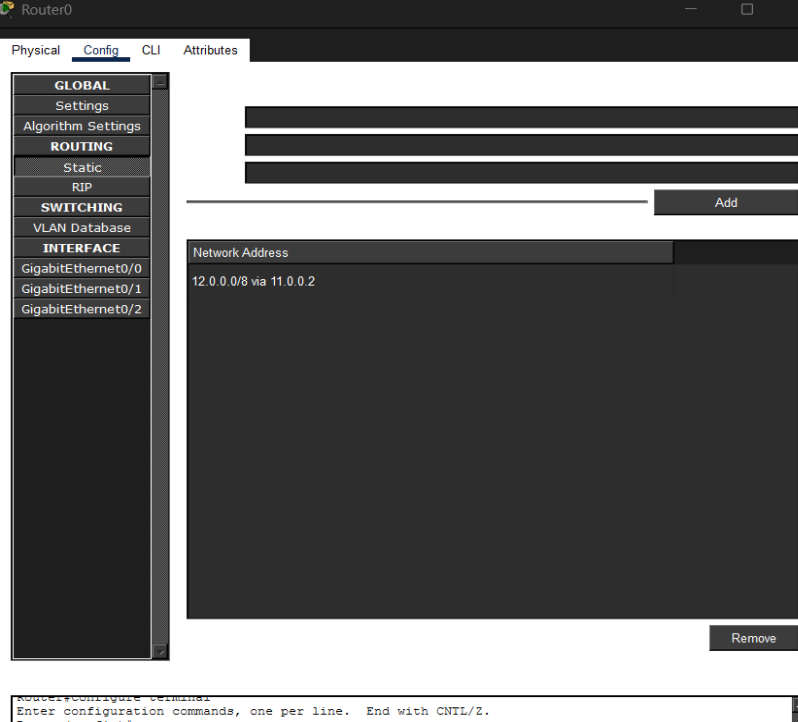
- enable
- configure terminal
- interface
- ip address
- no shutdown
- ip route
- show ip route
- show ip interface brief
- ping
- write memory

DEMONSTRATION/OUTPUT:

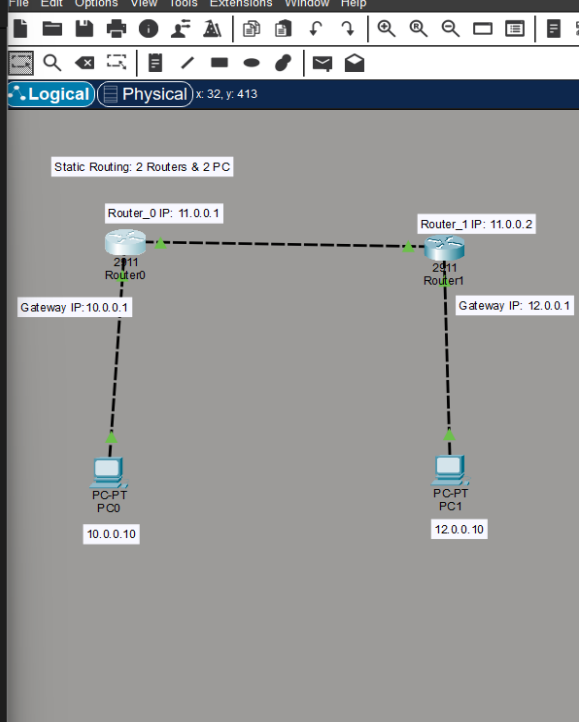
Part-1: 2 routers, 2pc's



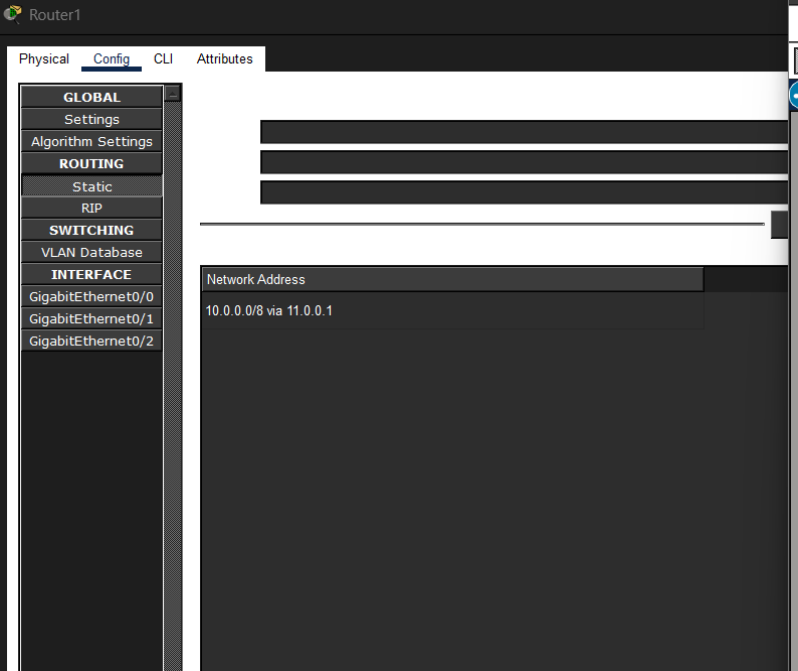
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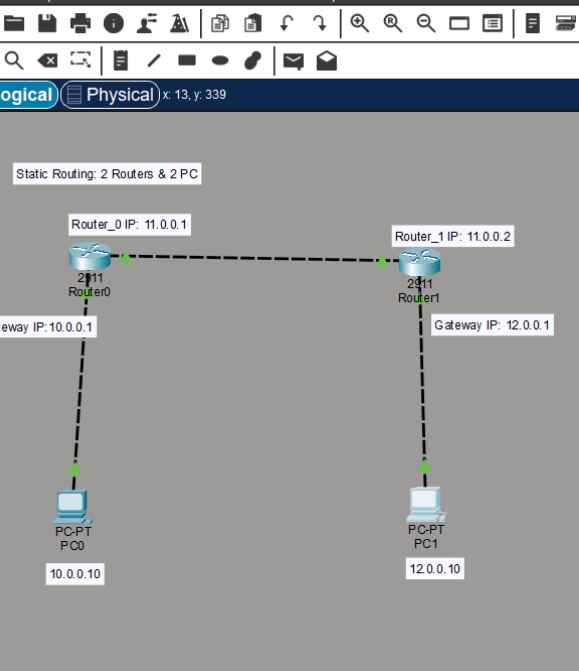
Router0 Configuration Screenshot: The 'Config' tab is active. Under the 'INTERFACE' section, 'GigabitEthernet0/1' is selected. The 'Network Address' field shows '12.0.0.0/8 via 11.0.0.2'. The 'CLI' tab shows the command prompt 'Router0#configure terminal'.



Router0 Logical Diagram Screenshot: Shows a static routing setup with two routers (Router_0 IP: 11.0.0.1 and Router_1 IP: 11.0.0.2) connected by a dashed line. Router_0 is connected to PC-PT PC0 (10.0.0.10) via a Gateway IP: 10.0.0.1. Router_1 is connected to PC-PT PC1 (12.0.0.10) via a Gateway IP: 12.0.0.1.

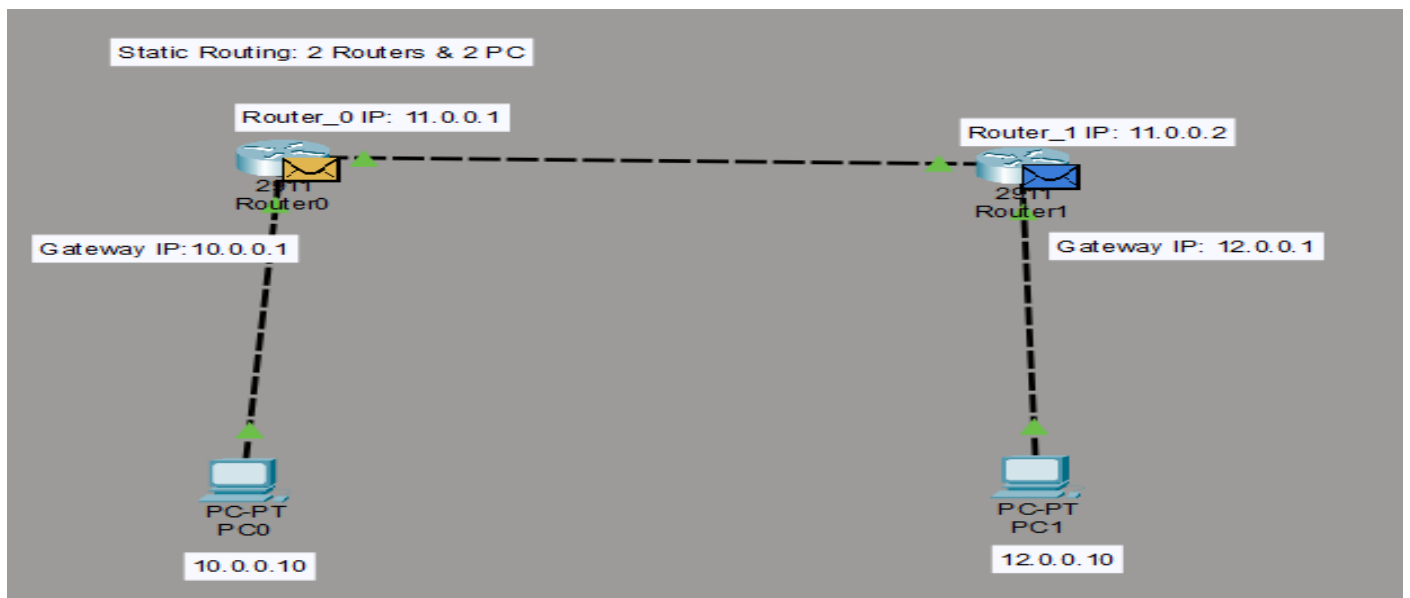
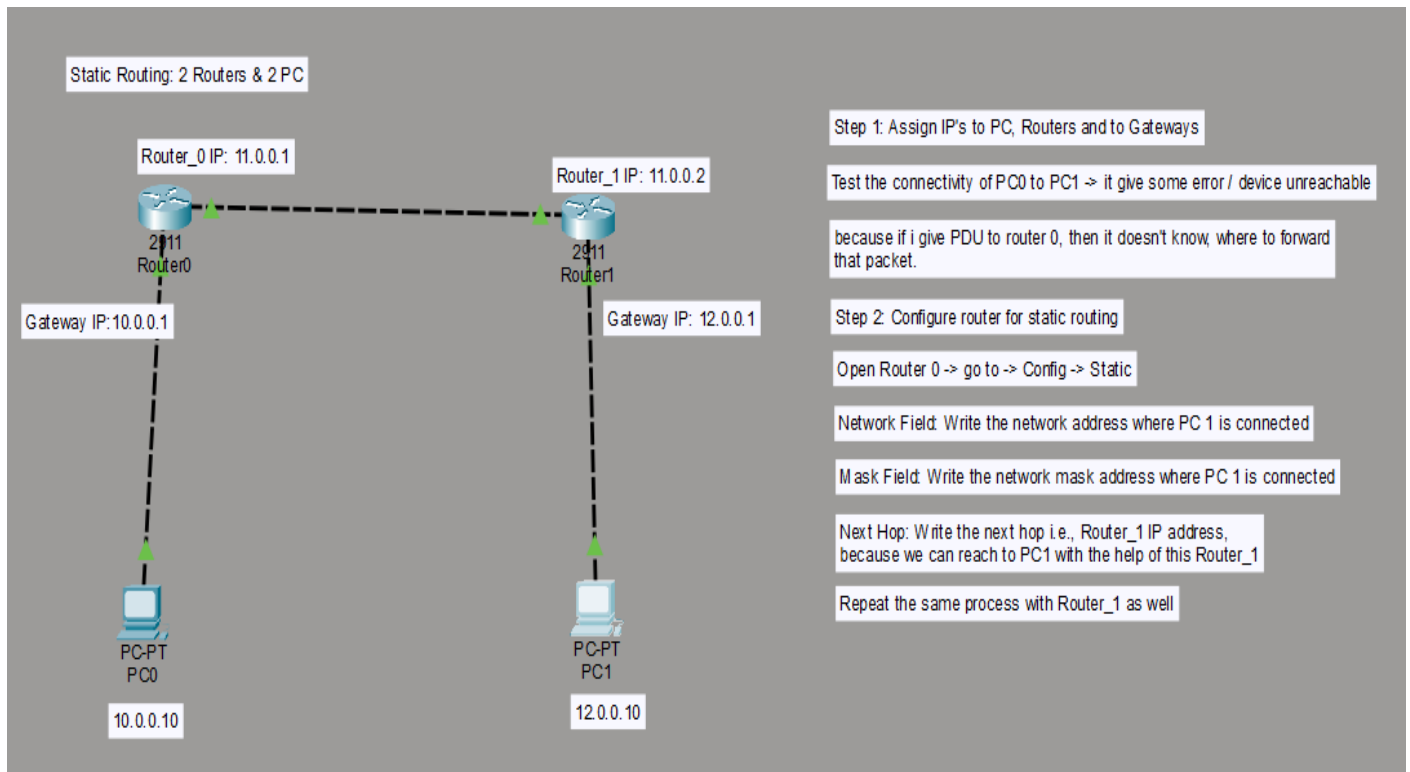


Router1 Configuration Screenshot: The 'Config' tab is active. Under the 'INTERFACE' section, 'GigabitEthernet0/1' is selected. The 'Network Address' field shows '10.0.0.0/8 via 11.0.0.1'. The 'CLI' tab shows the command prompt 'Router1#configure terminal'.



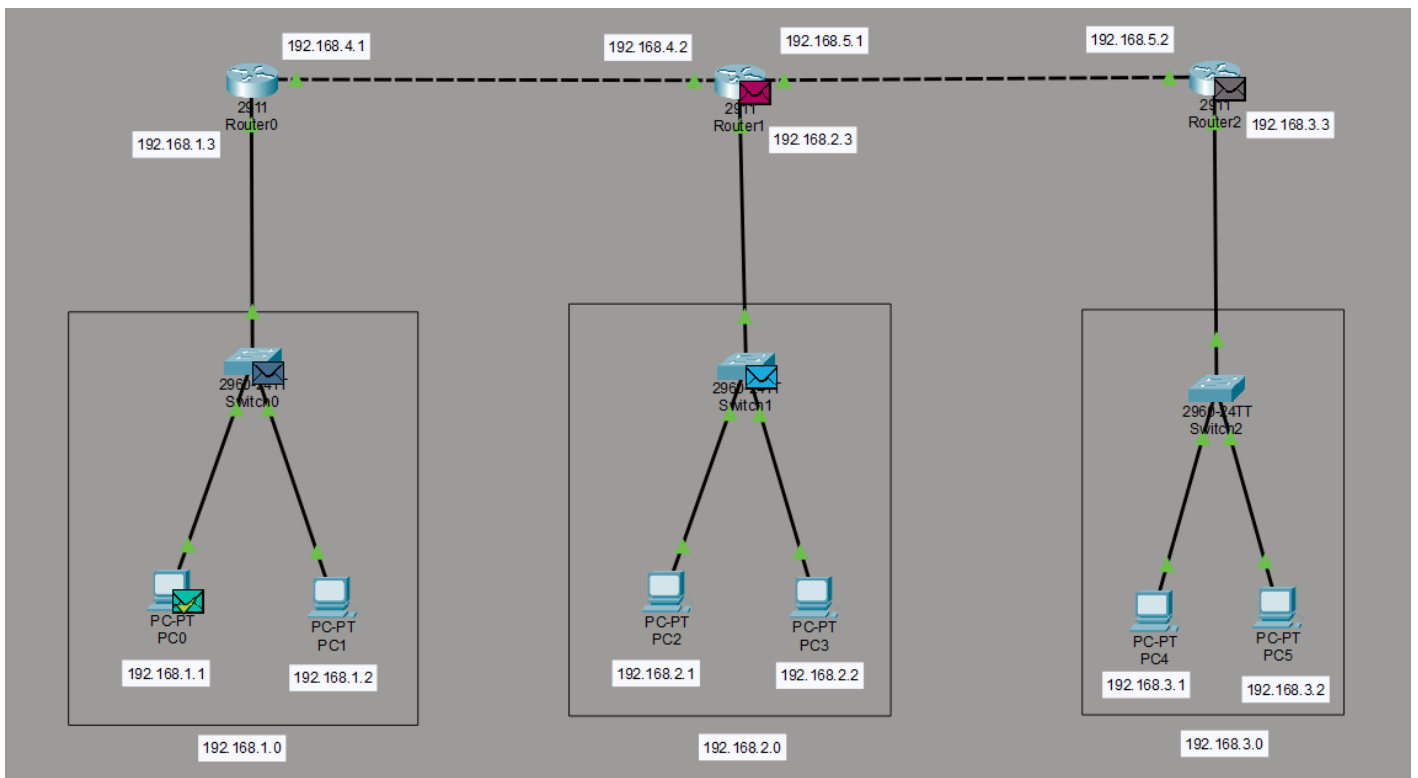
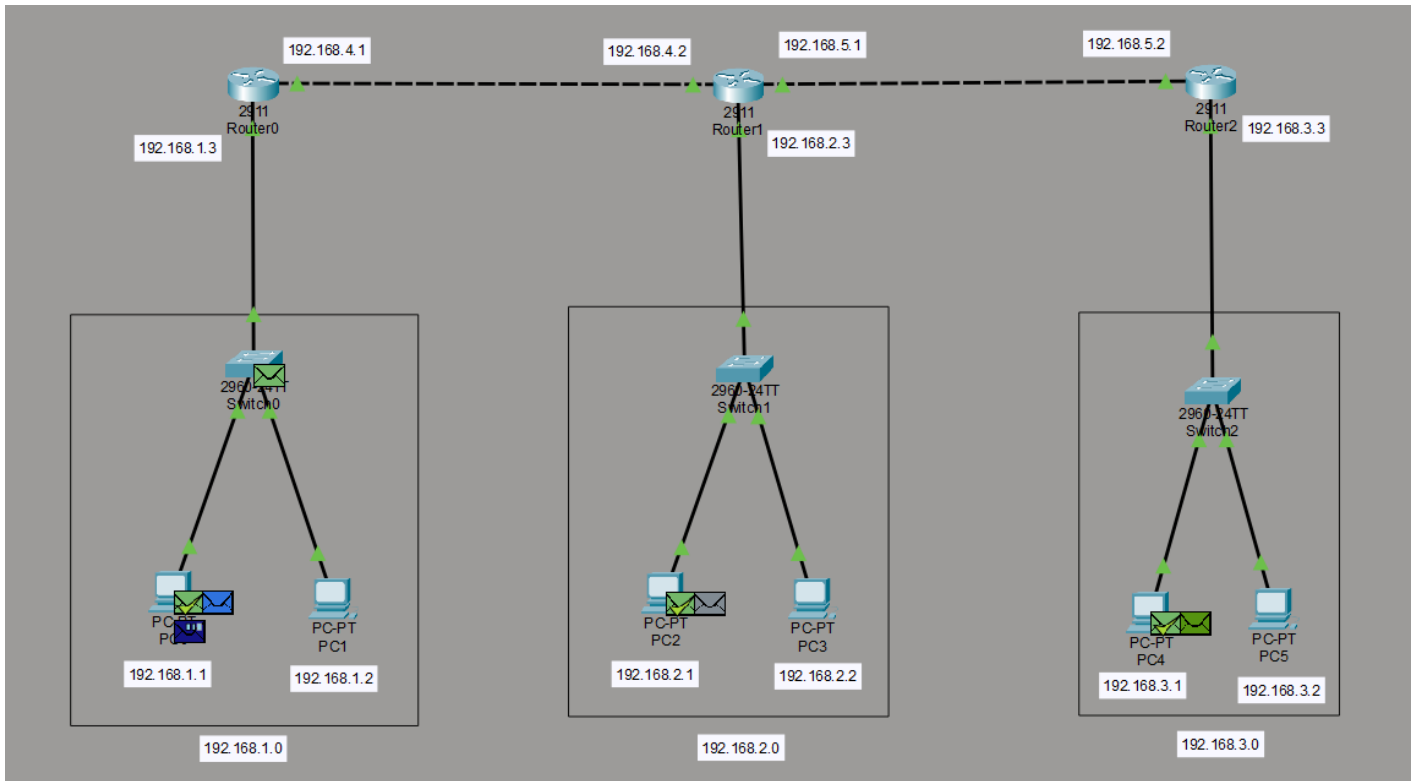
Router1 Logical Diagram Screenshot: Shows a static routing setup with two routers (Router_0 IP: 11.0.0.1 and Router_1 IP: 11.0.0.2) connected by a dashed line. Router_0 is connected to PC-PT PC0 (10.0.0.10) via a Gateway IP: 10.0.0.1. Router_1 is connected to PC-PT PC1 (12.0.0.10) via a Gateway IP: 12.0.0.1.

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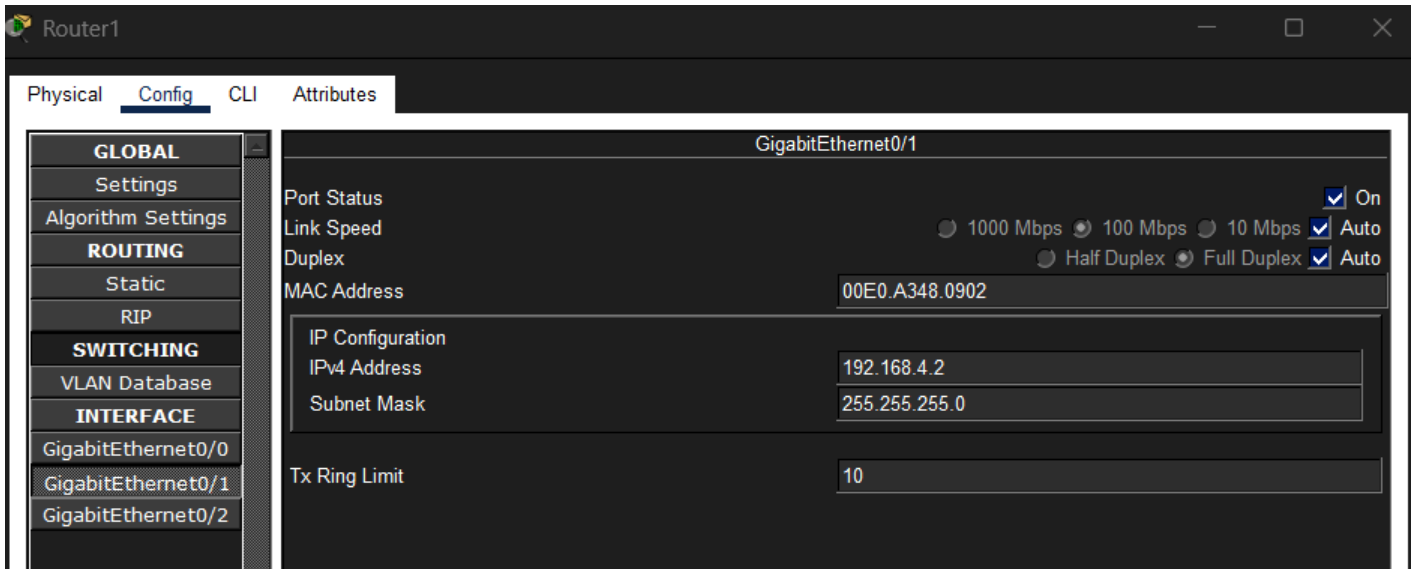


Part-2: 3 routers, 3 switches, 3pc's

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LEARNING OUTCOMES:

After completing this experiment:

- I understood the concept of routing and the importance of routers in enabling communication between different networks by forwarding packets based on IP addresses.
- I learnt to configure Static Routing by manually adding routes on routers using destination networks, subnet masks, and next-hop IP addresses, and verified the routing process through routing tables.
- I gained practical experience in configuring multiple LANs by assigning IP addresses, subnet masks, and default gateways to hosts, routers, and network interfaces using Cisco Packet Tracer.
- I developed the ability to design, configure, and troubleshoot routed network topologies involving two-router and three-router configurations, ensuring successful end-to-end communication between geographically separated networks.
- I compared Static Routing and Dynamic Routing based on configuration, scalability, maintenance, resource utilization, security, and adaptability, thereby understanding the practical scenarios where each routing technique is most suitable.